Robin Macharg

Senior / Principal iOS & Multi-Platform Software Engineer

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LinkedIn | GitHub | StackOverflow

Professional Summary

Security-cleared native mobile engineer *with extensive commercial experience delivering robust, high-performance iOS applications and SDKs using Swift, Objective-C, and C/C++.*

Proven track record in secure architectures *including VPN-secured Government communications (SC Cleared) and penetration-remediated enterprise applications.*

Deep expertise in modern development patterns *such as MVVM, Clean Architecture, Composable Architecture (TCA), and multi-platform native SDK design.*

End-to-end automation specialist *expert in App Store Connect, TestFlight, Fastlane, and high-efficiency CI/CD pipeline optimization.*

Accomplished collaborative lead *with a focus on Agile processes, mentoring junior developers, and scaling white-label platforms.*

Security-Focused High-Scale iOS Engineering Alignment

- **Native iOS Engineering & Tooling:** Proficient with Swift, Objective-C, Xcode, TestFlight, and App Store Connect, delivering multiple live apps to the Apple App Store.
- **Secure & Optimized Architecture:** Deep hands-on experience implementing MVVM, TCA, and Clean Architecture alongside low-level C++, cryptography, and secure interop.
- **Rigorous Penetration & Security Remediations:** Proven ability to implement secure device identity, VPN-secured local storage, SSO, and secure payment integrations.
- **Scalable Product Delivery:** Led white-label platforms scaling to 13 applications, driving over £30M+ in annual revenue with automated release processes.
- **Low-Level Debugging & Crash Analysis:** Extensive experience diagnosing complex tracebacks, memory leaks, and low-level system integrations for SDK environments.
- **Agile & Collaborative Workflows:** Certified Scrum Master possessing excellent track record working within international, multi-component development teams.

Education, Certifications & Credentials

- **BSc (Hons) Artificial Intelligence and Computer Science (Joint)** — University of Edinburgh
- Certified Scrum Master (CSM)

Selected Key Roles

Lead Mobile Developer | Boostworks Ltd.

November 2023 - Present

Keywords: [React Native, TypeScript, iOS/Android, CI/CD (Fastlane/Azure), Apple/Google Pay, SSO, Penetration Testing]

- Commercial Ownership & Scale: Solely maintained a 60K-line multi-tenant codebase driving £30M+ annual revenue across 13 white-label iOS/Android apps for flagship enterprise clients.
- Platform Security & Enterprise Payments: Led native component wrapping (Swift/Kotlin) for secure Apple/Google Pay integrations, executed penetration testing remediations, and integrated enterprise SSO.
- Automation & CI/CD: Built end-to-end pipelines (Fastlane, Azure, OpenAPI) to slash new application deployment times by 90%.

Senior iOS Developer | Sky

October 2021 - April 2022

Keywords: [Swift, Kotlin Multiplatform, SwiftUI, Video Streaming SDKs, Agile, Release Automation]

- High-Scale SDK Management: Maintained a hybrid Swift and Kotlin Multiplatform Mobile (KMM) SDK and reference apps utilized daily by over 25 million customers.

- Engineering Improvement: Rewrote the internal SDK reference application using SwiftUI, ensuring highly optimized performance and clean architecture.
- Release & Delivery: Managed bi-weekly release cycles, release notes, and custom Probot/Octokit GitHub tools for repo automation.

Senior Apple Engineer | Bugsnag Inc.

January 2020 - June 2020

Keywords: [Objective-C, C++, iOS, macOS, Crash Reporting, Open Source, Diagnostics]

- Low-Level Diagnostics & Cocoa/C++: Maintained high-performance open-source Bugsnag Cocoa and C/C++ diagnostic libraries serving global enterprise clients like Netflix, Nintendo, and Amazon.
- Root Cause Engineering: Conducted deep system diagnostics, handling complex multi-threaded stack traces, crash log analysis, and targeted patch releases.
- Distribution Engineering: Managed library packaging across CocoaPods, Carthage, SPM, and manual installations.

Principal iOS Developer | Amiosec Ltd. (Contract)

March 2019 - December 2019

Keywords: [Xamarin, C#, Objective-C/Swift Interop, Security, SSL MiM, EWS, Foxhound Project]

- Secure Defense Communications: Engineered a custom VPN-secured Exchange email client for the UK Government's secure Foxhound project.
- Secure Interoperability: Developed robust, secure interop bindings between Xamarin/C# and Objective-C/Swift utilizing .NET embedding.
- Advanced Network Inspection: Reverse-engineered Exchange Web Services (EWS) using secure SSL Man-in-the-Middle proxying via Charles Proxy.
- Modern Reactive Patterns: Implemented MVVM architecture leveraging ReactiveUI, DynamicData, and Realm.NET with robust error handling via Polly.

Principal iOS Developer | Dyson Ltd. (Contract)

June 2017 - January 2020

Keywords: [iOS, Enterprise MDM, ARKit, RealmDB, Swift, Unit Testing, Penetration Testing]

- Enterprise iOS Engineering: Designed and delivered Dyson's first B2B sales iPad app ('Hero') for global deployment to hundreds of staff.
- Secure Enterprise Integration: Managed end-to-end integration covering AirWatch enterprise MDM deployment, extensive penetration testing, and secure CMS backends.
- Robust Architecture & Testing: Achieved over 80% UI and Unit Test coverage utilizing RealmDB and custom Xcode project templates.

Earlier Career Highlights

2012 - 2023 Broad engineering roles spanning private equity fintech platforms, mapping SDKs, startup growth, and high-performance network analytics.

- **Senior Developer** | *Floww Ltd* (iOS, SwiftUI, TCA, Redux, CI/CD, Git hooks)
- **Lead Mobile Developer** | *This One Ltd* (SwiftUI, Combine, Async/Await, Python, Flask, Fastlane, GCP)
- **Senior Developer** | *Living Map Ltd* (iOS, XCFrameworks, CocoaPods, Carthage, Background Location)
- **Head of Mobile Development** | *Actual Experience Ltd.* (Objective-C, Swift, CoreData, Realm, Low-level Networking (ICMP/TCP))
- **Principal Mobile Developer** | *Accident Exchange Group (Contract)* (iOS, CoreData, CoreLocation, MapKit, RestKit)
- **Senior Middleware & DevOps Developer** | *MySpace* (Java, Python, JBoss, Puppet, Objective-C)
- **Senior Developer / Engineer** | *RMS Services Ltd / Airbus UK* (Python, PostgreSQL, PyQt, C/C++, OpenGL)
- **Server & Embedded Developer** | *Sift / Appliance Studio / Obtree* (Linux, Embedded Systems, Java, Perl, C)

Open Source Contributions

- Bricksmith: SourceForge award-winning LEGO CAD software for macOS (Objective-C/C).
- Pycel: Python-based Excel dependency graph and compiler formula parser.
- React Native Input Spinner: Regular contributor to popular open-source UI component.
- RMHelpfulLabel: Open-source custom iOS UI helper components and utility libraries.
- Coding Exercises: Public repository of algorithms and structured programming solutions.

Technical Skills and Expertise

Native iOS Ecosystem & Languages - Swift, Objective-C, SwiftUI, Combine, async/await, UIKit, CoreData, RealmDB, XCFrameworks

Security & Architecture - VPN integrations, Penetration testing remediation, SSO, Apple/Google Pay, MVVM, TCA, Clean Architecture, Scrum (CSM)

Tooling & Delivery - Xcode, TestFlight, App Store Connect, Fastlane, Git, GitHub Actions, Azure Pipelines, xcodebuild, SPM, CocoaPods, Carthage

Multi-Platform & Systems - C/C++, Python (FastAPI, Flask), Kotlin Multiplatform (KMM), C#, Docker, Shell Scripting